

ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

NER AND TOKENIZATION

UC San Diego

COMPUTER SCIENCE & ENGINEERING
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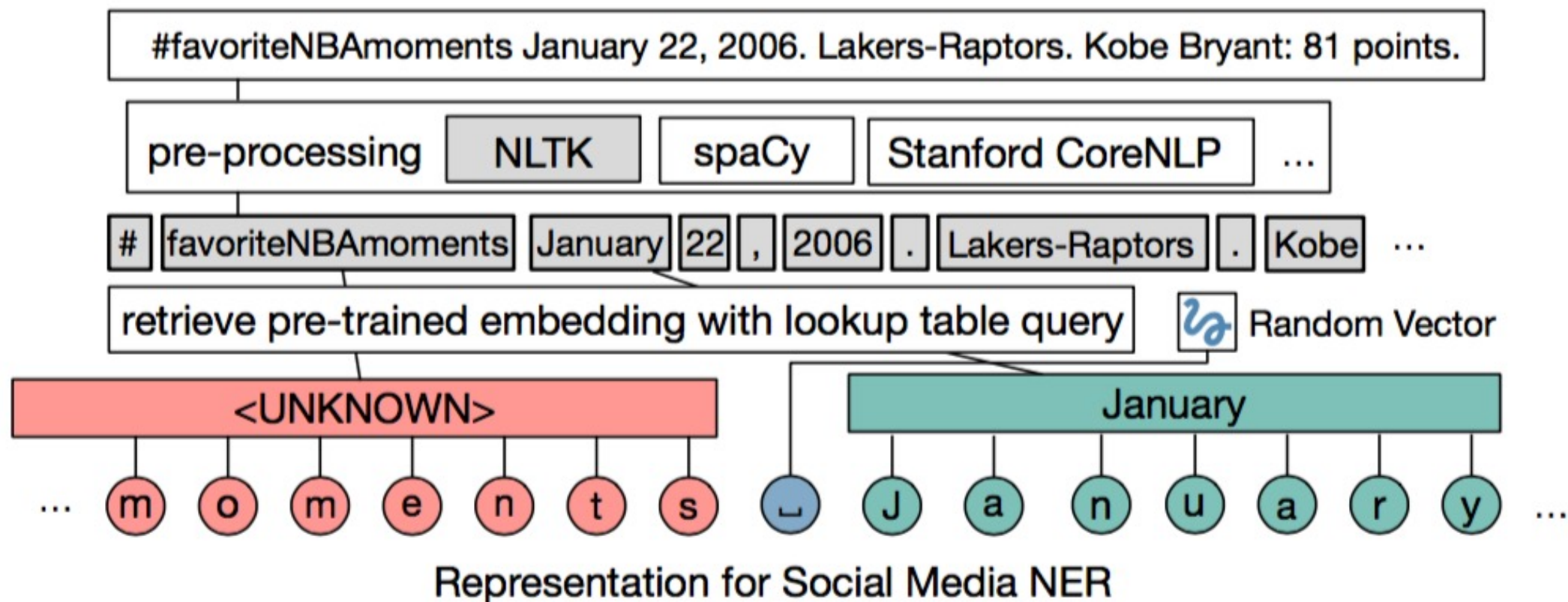
Classic Named Entity Recognition (NER)

Some questions to think about:

- Can massive raw texts help? → Neural language model
- Can dictionaries help? → Distant supervised setting
- Are human annotations always correct? → Auto-Correction
- **Is Tokenizer always good?**

TYPICAL NER PIPELINE SYSTEM

Pre-processing tools are applied first.



Broad Twitter Corpus (BTC)

- A twitter NER dataset

spaCy

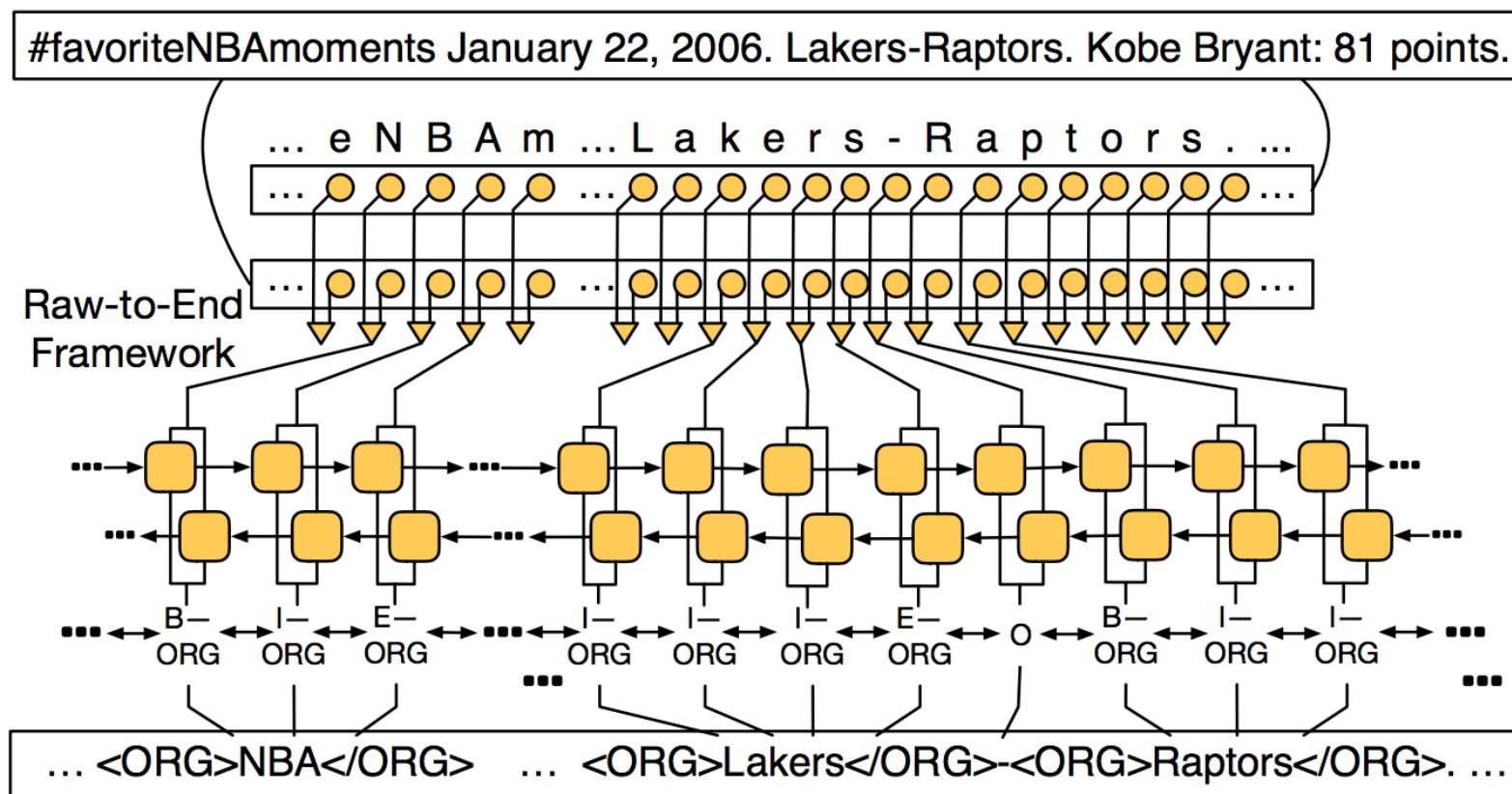
- A popular Python NLP lib

spaCy tokenization + BTC dataset

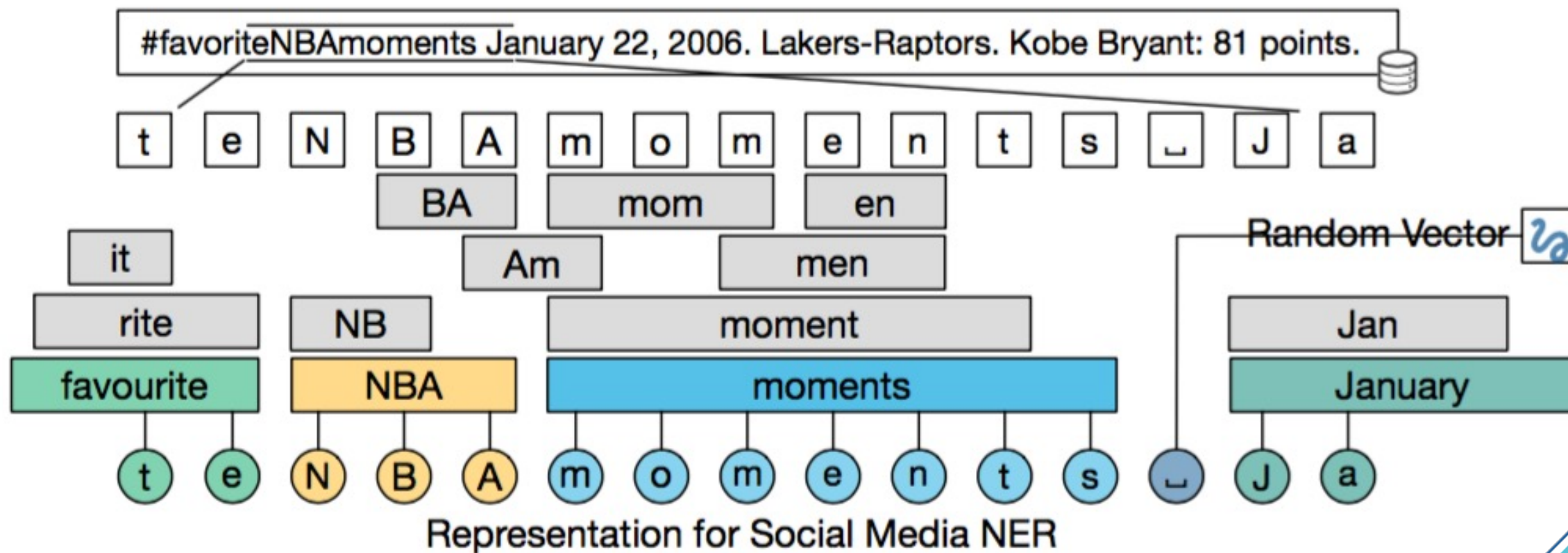
- → Word boundaries of more than **45%** named entities will be incorrectly identified!

We propose to conduct NER training in a raw-to-end manner

- Raw text as the input & Predictions at the character level



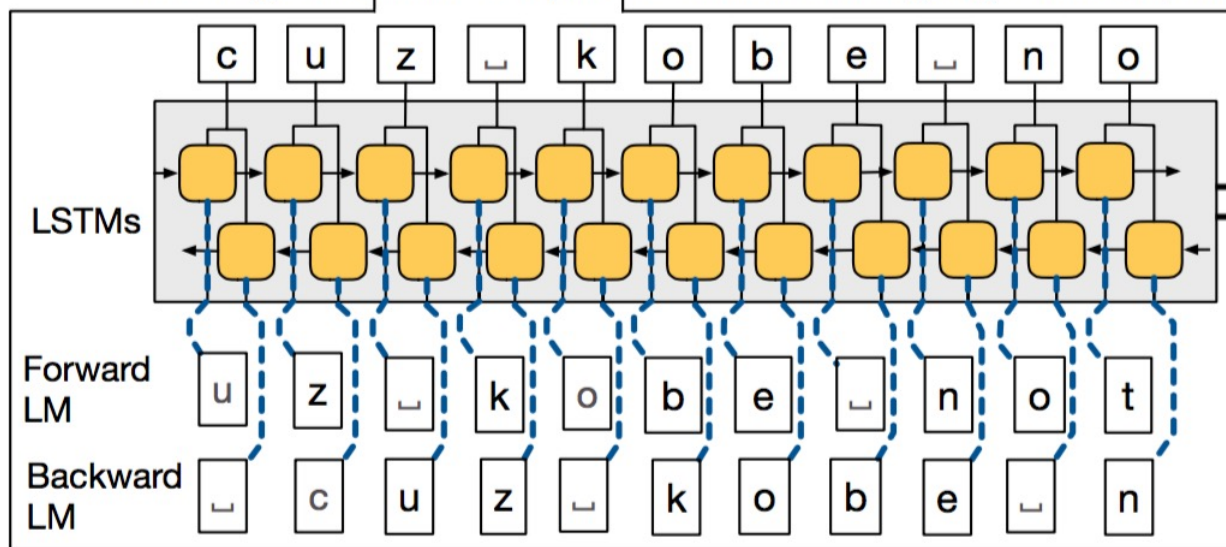
Prefer to match the words with higher Inverse Document Frequency (IDF)



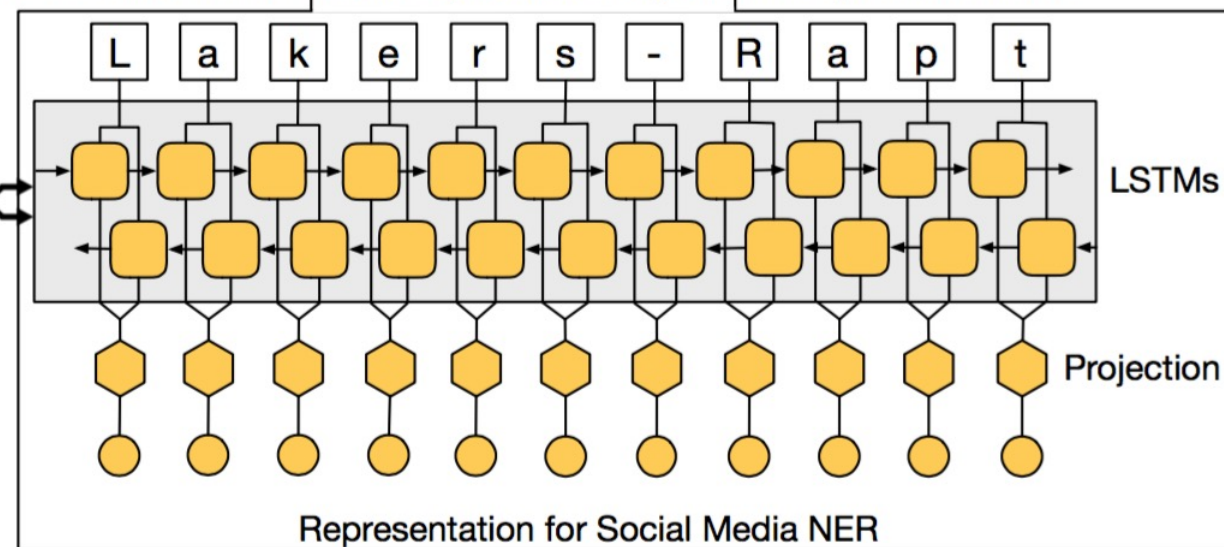
Character-level neural language model is leveraged

- Pre-training + Contextualized representations

Leverage the **Raw Corpus** to construct language models, which are used in the **Task-Specific Corpus** for representations



(a) Language Model Pre-training



(b) Integrated as the Contextualized Character Representation Model

Tokenizer matters

- NLTK is the best on both datasets

Raw-to-End wins

- String Match is even better

Tokenizer	Methods	TNT	BTC
NLTK	TwitterNER	73.41	64.40
	LSTM-CNN-CRF	80.25	66.48
	LM-LSTM-CRF	80.85	67.73
	Flair	83.26	68.33
spaCy	TwitterNER	70.40	37.21
	LSTM-CNN-CRF	71.66	29.04
	LM-LSTM-CRF	72.25	31.17
	Flair	73.49	33.20
Stanford CoreNLP	TwitterNER	70.18	38.01
	LSTM-CNN-CRF	64.22	29.71
	LM-LSTM-CRF	63.14	28.97
	Flair	65.58	32.39
Neural-Char-CRF (Match)		86.96	74.98
Neural-Char-CRF (NLTK)		85.59	74.39

- Using neural language model, massive raw texts can help!
- High-quality dictionaries can help!
- Human annotations are NOT always correct!
- Tokenizer is not that important and sometimes even hurts!

RECOMMENDED READINGS

- Lample, Guillaume, et al. "Neural Architectures for Named Entity Recognition." NAACL 2016.
- Ma, Xuezhe, and Eduard Hovy. "End-to-end Sequence Labeling via Bi-directional LSTM-CNNs-CRF." ACL 2016.
- Liu, Liyuan, et al. "Empower sequence labeling with task-aware neural language model." AAAI 2018.
- Liu, Liyuan, et al. "Efficient Contextualized Representation: Language Model Pruning for Sequence Labeling." EMNLP 2018.
- Fries, Jason, et al. "Swellshark: A generative model for biomedical named entity recognition without labeled data." arXiv preprint arXiv:1704.06360 (2017).
- Shang, Jingbo, et al. "Learning Named Entity Tagger using Domain-Specific Dictionary." EMNLP 2018.
- Wang, Zihan, et al. "CrossWeigh: Training Named Entity Tagger from Imperfect Annotations." EMNLP 2019.
- Liu, Liyuan, et al. "Raw-to-end name entity recognition in social media." arXiv preprint arXiv:1908.05344 (2019).